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A Markov Model of Switching-Regime ARCH

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In this article I present a new approach to model more realistically the variability of financial time series. I develop a Markov–ARCH model that incorporates the features of both Hamilton's switching-regime model and Engle's autoregressive conditional heteroscedasticity (ARCH) model to examine the issue of volatility persistence in the monthly excess returns of the three-month treasury bill. The issue can be resolved by taking into account occasional shifts in the asymptotic variance of the Markov–ARCH process that cause the spurious persistence of the volatility process. I identify two periods during which there is a regime shift, the 1974:2–1974:8 period associated with the oil shock and the 1979:9–1982:8 period associated with the Federal Reserve's policy change. The variance approached asymptotically in these two episodes is more than 10 times as high as the asymptotic variance for the remainder of the sample. I conclude that regime shifts have a greater impact on the properties of the data, and I cannot reject the null hypothesis of no ARCH effects within the regimes. As a consequence of the striking findings in this article, previous empirical results that adopt an ARCH approach in modeling monthly or lower frequency interest-rate dynamics are rendered questionable.

KEY WORDS: Asymptotic variance; Autoregressive conditional heteroscedasticity; Switching-regime.

1. INTRODUCTION

One of the interesting issues of financial time series volatility relates to the persistence of shocks to the variance. A common finding using high-frequency financial data concerns the apparent persistence implied by the estimates for the conditional variance functions. As a result, Bollerslev and Engle (1986) introduced the class of integrated generalized autoregressive conditional heteroscedasticity (GARCH) models, or IGARCH. The IGARCH has the property called "persistence in conditional variance," which means that the current information remains important for the forecasts of the conditional variances for all future horizons.

The test for IGARCH is a test for unit roots in the variance. Formal tests for a unit root in the variance have been performed by several authors, and the null hypothesis of a unit root is typically not rejected. See, for example, French, Schwert, and Stambaugh (1987), Chou (1988), Schwert and Seguin (1990), and Bollerslev, Chou, and Kroner (1992). The ARCH and GARCH models for interest rates typically exhibit parameters that are not in the covariance stationary region. For example, Hong (1988) found a unit root in the excess return of the three-month treasury-bill (T-bill) rate over the one-month T-bill rate, and Engle, Lilien, and Robins (1987) provided more striking results on the explosive parameter of their ARCH-MEAN model for the excess holding return on the six-month T-bill over the three-month T-bill.

Lamoureux and Lastrapes (1990) pointed out that one potential problem with IGARCH is that, unlike the random walk in the mean of asset prices, it lacks theoretical motivation. The idea of a drift in the variance is intuitively appealing, but it is difficult to justify by an analogy with models of a drift in mean. Lamoureux and Lastrapes (1990) formally investigated the possibility that the appearance of IGARCH is due to time-varying GARCH parameters and found that allowing for the presence of deterministic shifts in the conditional variance intercept in GARCH produces substantially lower estimates of the persistence parameters for 30 randomly selected Center for Research in Security Prices (CRSP) stocks. Diebold (1986) also indicated that although interest-rate equations appear to have integrated-variance disturbances, it may be due to the failure to include a monetary-regime dummy for the conditional variance intercept. This will correspond to a convariance-stationary GARCH movement within regimes, with a "jump" occurring between regimes. It is this suggestion that is formally addressed in this article.

One possible way to endogenize changes in the intercept of an ARCH model is to assume that the intercept follows a jump process. This is the approach taken by Jorion (1988), who investigated the existence of discontinuities in the sample path of exchange rates and the sample path of a value-weighted stock-market index. The main difficulty with this approach is its empirical tractability. Although Ball and Torous (1985) showed that in practice one needs to truncate the log-likelihood function for one observation and usually truncation at 10 provides satisfactory accuracy, it is still quite cumbersome.

On the other hand, Hamilton (1988) proposed a Markov-switching model for capturing the effects of sudden dramatic political and economic events on the properties of financial and economic time series. The model is especially suitable for a situation like the 1979:10–1982:10 Federal Reserve experiment, which leads not only to unprecedented interest-rate levels but also to unprecedented interest-rate volatility. The apparent success of the switching-regime model with short-term interest rates was clearly documented

by Hamilton (1988) and Driffill (1992). Other applications include Hamilton's (1989) model of long-run trends in gross national product and the business cycle, Turner, Startz, and Nelson's (1989) model of excess return and volatility, Cecchetti, Lee, and Mark's (1990) explanation of mean reversion in equilibrium asset prices, and Engel and Hamilton's (1990) model of exchange-rate dynamics.

As appealing as the Markov-switching model is, it may not capture all the movements in the variance. For example, Pagan and Schwert (1990) compared several statistical models for monthly U.S. stock-return volatility from 1834 to 1925 and found that Hamilton's model provides only a weak explanation of the data. In the case of monthly short-term interest rates, Hamilton (1988) concluded that both regime-shift and ARCH effects seem to be present in the data, though the regime shift is by far the most dramatic source of conditional heteroscedasticity over the full sample period. Hamilton (1988) also suggested that the regime shift may account for the explosive parameter estimates in the ARCH-MEAN model of Engle et al. (1987).

The purpose of this article is to combine the Markov-switching model of Hamilton with the ARCH model of Engle (1982) to describe financial time series. I model explicitly the dynamics of the asymptotic variance in the switching-regime ARCH (the value that the variance would approach asymptotically on a long but finite stretch of a realization of the Markov chain that remained in the same state) as following a first-order Markov process. In this case, the conditional variance is no longer determined by an exact linear combination of past conditional variances and past shocks as in the standard ARCH(g) process. The intercept in the conditional variance, however, need not change every period but only in response to occasional discrete events. Thus the model should be able to retain the volatility-clustering feature of the ARCH model and, in addition, capture the discrete shift in the intercept in the conditional variance that may cause spurious apparent persistence in the variance process. Combining a switching-regime model with an ARCH model endogenizes the parameter shifts and lets the data decide at what time a sudden dramatic event has occurred. In addition, it enables us to distinguish which effect is more important in terms of its impact on the properties of the data.

Unfortunately, combining the Markov-switching model with GARCH induces tremendous complications in actual estimation. It can be shown that as a result of the particular lag structure of the GARCH model, the normal equations from the EM algorithm for the parameter estimation depend on the entire history of the state through the smoothed probabilities $p(S_T, \ldots, S_1 \mid I_T)$, where S_t denotes the state at time t and I_T refers to the time T information set. Because each of the S_t's takes on two values, this implies a total of 2^T probabilities that need to be managed. Any data series with a sample size larger than 50 will consequently be extremely difficult to estimate. Direct maximum likelihood estimation via a nonlinear filter is also practically infeasible. We therefore restrict the dynamic lag structure to an ARCH model in the empirical analysis of the monthly excess returns of the three-month T-bill.

In related independent work, Hamilton and Susmel (1992) proposed another parameterization of the switching-regime ARCH (SWARCH) model. The idea of their SWARCH model is to model changes in regime as changes in the scale of the ARCH process. For the weekly returns of the New York Stock Exchange value-weighted index, they find that the SWARCH specification offers a better statistical fit to the data, better forecasts, and a better description of the October 1987 crash. Their estimates attribute most of the persistence in the stock-price volatility to the persistence of the low-, moderate-, and high-volatility regimes, which typically last for several years. The high-volatility regime is associated with economic recessions. In addition, they find that only two lags in the ARCH process are needed to describe the volatility process and that the persistence measure of the ARCH process is much lower.

The article is organized as follows: Section 2 describes the theoretical model, Section 3 presents the empirical results, and Section 4 concludes.

2. THE SWITCHING–AR(K)–MARKOV–ARCH(G) MODEL

I postulate the following model of excess return series, r_t:

$$
\begin{aligned}
r_t &= c_0 + c_1 S_t + z_t \\
z_t &= b_1 z_{t-1} + \cdots + b_k z_{t-k} + \epsilon_t \\
\epsilon_t &= u_t \sqrt{h_t}, u_t \, i.i \sim N(0,1) \\
h_t &= \gamma(S_t) + \sum_{i=1}^{g} \alpha_i \epsilon_{t-i}^2, \qquad \alpha_i \geq 0,
\end{aligned} \tag{1}
$$

where

$$
\gamma(S_t) = \gamma_0 + \gamma_1 S_t, \qquad \gamma_0 > 0, \gamma_1 > 0, \tag{2}
$$

and z_t is the deviation from the regime mean $c_0 + c_1 S_t$, where $S_t = 0$ or 1 denotes the unobserved state of the system. The latent variable S_t is assumed to follow a first-order Markov process with the transition probabilities

$$
\begin{aligned}
p[S_t = 0 \mid S_{t-1} = 0] &= p \\
p[S_t = 1 \mid S_{t-1} = 0] &= 1 - p \\
p[S_t = 1 \mid S_{t-1} = 1] &= q \\
p[S_t = 0 \mid S_{t-1} = 1] &= 1 - q.
\end{aligned} \tag{3}
$$

The stochastic process for S_t is strictly stationary if both p and q are less than unity and do not take on the value of 0 simultaneously. It admits the following (autoregressive) AR(1) representation:

$$
S_t = (1 - p) + \eta S_{t-1} + V_t, \qquad \eta = -1 + p + q. \tag{4}
$$

Equation (4) is a special case of a standard AR(1) model, with an unusual probability distribution of the innovation sequence $\{V_t\}$. As shown by Hamilton (1988), η can be interpreted as a measure of persistence in the forcing process.

The preceding specification allows the structural changes in both the mean and the constant intercept in the ARCH

specification. The system is said to be in a low-variance state when $S_t = 0$ and in a high-variance state when $S_t = 1$.

Hamilton and Susmel (1992) employed the related parameterization in their SWARCH model:

$$\epsilon_t = \sqrt{\gamma_{S_t}}\tilde{\epsilon}_t, \quad \tilde{\epsilon}_t = u_t\sqrt{h_t}, \qquad u_t \sim i.i.N(0,1),$$
$$h_t = \alpha_0 + \alpha_1\tilde{\epsilon}_{t-1}^2 + \alpha_2\tilde{\epsilon}_{t-2}^2 + \cdots + \alpha_g\tilde{\epsilon}_{t-g}^2,$$

where ϵ_t is the innovation as described in Equation (1). With this specification, the scaled residual $\tilde{\epsilon}_t$ follows a standard ARCH(g) process. The process is therefore multiplied by the constant $\gamma_1^{1/2}$ when the process is in the regime represented by $S_t = 1$, multiplied by $\gamma_2^{1/2}$ when $S_t = 2$, and so on. I shall not pursue this specification any further, however.

Sufficient condition for the process $\{\epsilon_t\}$ defined by Equations (1), (2), and (3) to be covariance stationary is given by the following proposition.

Proposition. Assume that $\{\epsilon_t\}$ starts indefinitely far in the past with finite variance; then $\{\epsilon_t\}$ will be covariance stationary if $\eta = -1 + p + q < 1$ and the roots of the equation

$$1 - \sum_{i=1}^{g} \alpha_i x^i = 0 \tag{5}$$

are all outside the unit circle, where α_i are the ARCH coefficient. If $\{\epsilon_t\}$ is covariance stationary, the unconditional variance is given by

$$E(\epsilon_t^2) = \frac{\gamma_0 + \pi\gamma_1}{1 - \sum_{i=1}^{g}\alpha_i},$$

where $\pi = (1 - p)/(2 - p - q)$ corresponds to the unconditional probability of being in state 1.

Proof. See the Appendix.

Note that the sufficient and necessary condition for the stationarity of the standard ARCH process is that the roots of Equation (5) are all outside the unit circle. With the additional Markov structure, we need also for η to be smaller than unity.

3. THE EMPIRICAL RESULTS

3.1 The Data

The data used in this article are the monthly excess returns of the three-month T-bill from August 1964 to November 1991. The monthly excess return is constructed by subtracting the one-month T-bill rate, under the assumption that it represents a risk-free return, from the monthly return on the three-month T-bill. The data are from the CRSP Fama holding period return file as given by Engle, Ng, and Rothschild (1990). In their article, they plotted the monthly excess returns and squared monthly excess returns for the 2-, 4-, 6-, 8-, 10-, and 12-month T-bills and the the cross-products of the three-month T-bill excess returns with the excess returns of these assets. They found common periods of high volatility, especially during the 1973–1975 and 1979–1983 periods. Although the empirical results presented later pertain directly to the three-month T-bill, I believe that the results are robust for. T-bills of different maturities. Figure 1 plots the three-month

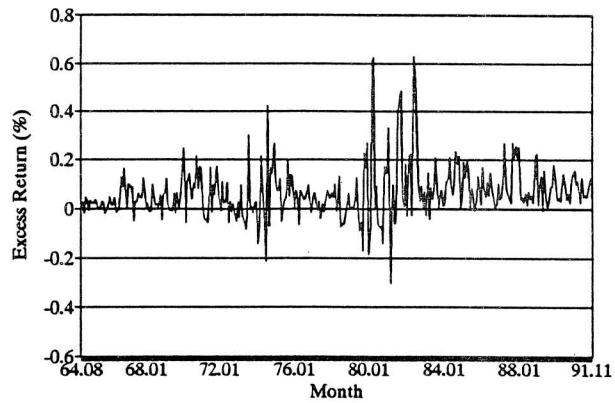


Figure 1. Three-Month Treasury-Bill Excess Return: 08/64–11/91.

T-bill percentage excess returns. The mean and the standard deviation of the series are .07 and .11, respectively. The skewness and kurtosis are 1.74 and 9.56 Ljung–Box statistics for the 20-order serial correlation in the levels and squares are 72.57 and 173.57, respectively, suggesting significant serial correlations for both the levels and the squares of the excess return series.

3.2 Maximum Likelihood Estimates for the AR(k)–Markov–ARCH(g) Model

The maximum likelihood estimates for the model in Equations (1)–(3) can be obtained via the EM algorithm or a nonlinear filter. Instead of evaluating the log-likelihood function directly, the EM algorithm iterates on the expectation of the log-likelihood function with respect to the distribution of the latent variable, the state variable S_t. The derivations of the EM algorithm for the Markov–ARCH model are provided in an earlier version of the article. Because I also allow for structural changes in the mean of the return series, the EM algorithm is extremely difficult numerically as a result of the large number of states I need to keep track of. I therefore employ a nonlinear filter to obtain the estimates. The details of these two approaches are available from me.

The switching–AR(k)–Markov–ARCH(g) model is estimated for $k = 1, 2, 3, 4$ and $g = 1, 2, 3$. In all cases, the third ARCH coefficient is essentially 0 and the likelihood-ratio test statistic provides no evidence against the hypothesis that it is 0. Therefore I choose $g = 2$. I then apply the Schwarz criterion to determine the lags in the autoregression. The Schwarz criterion chooses k to maximize

$$\frac{\ln L}{T} - \frac{1}{2}\frac{\ln T}{T}k,$$

where $\ln L$ is the value of the maximized log-likelihood function. The corresponding Schwarz values are 1.05, 1.04, 1.03, and 1.02 for $k = 1, 2, 3, 4$. I therefore choose $k = 1$.

The maximum likelihood estimates for the switching–AR(1)–Markov–ARCH(2) model are reported in Table 1. The mean excess return in state 0 (c_0) and state 1 ($c_0 + c_1$) are .06% and .12%, respectively. The AR(1) slope coefficient is .28 with a significant t statistic. For the variance, the first and second ARCH coefficients (α_1, α_2) have been greatly reduced to .06 and .09 with insignificant t statistics.

Table 1. A Comparison of Alternative Models for Three-month T-Bill Excess Return (1964:8–1991:11)

	Estimates					
Parameter	*Unrestricted*[a]	$c_1 = \gamma_1 = 0$	$\alpha_1 = \alpha_2 = 0$	$c_1 = 0$	$\gamma_1 = 0$	*GARCH*[b]
c_0	.0574	.0595	.0570	.0595	.0557	.0615
	(.0064)[c]	(.0059)	(.0059)	(.0060)	(.0049)	(.0064)
c_1	.0576		.0712		.4367	
	(.0419)		(.0384)		(.0331)	
b_1	.2830	.2167	.2726	.2926	.1665	.2769
	(.0599)	(.0660)	(.0589)	(.0571)	(.0637)	(.0615)
γ_0	.0036	.0035	.0040	.0036	.0037	.0004
	(.0006)	(.0007)	(.0004)	(.0005)	(.0005)	(.0003)
γ_1	.0346		.0391	.0368		
	(.0106)		(.0084)	(.0103)		
α_1	.0597	.4122		.0601	.3689	.2323
	(.0712)	(.1372)		(.0604)	(.1151)	(.0636)
α_2	.0854	.4021		.0867	.1454	
	(.0642)	(.1674)		(.0619)	(.0839)	
p	.9878		.9862	.9878	.9895	
	(.0094)		(.0090)	(.0094)	(.0061)	
q	.9402		.9372	.9480	.6008	
	(.0395)		(.0374)	(.0347)	(.1748)	
β						.7482
						(.0657)
Likelihood value	339.27	319.21	337.28	338.31	335.17	327.35
Likelihood ratio test		40.12	3.98	1.92	8.20	
		(.00)[d]	(.14)[e]	(.17)[f]	(.00)[f]	
Schwarz values[g]		1.05	1.04	1.03	1.02	
		($k = 1$)	($k = 2$)	($k = 3$)	($k = 4$)	

[a] The unrestricted model is described in Equation (1).
[b] The model is AR(1)–GARCH(1, 1) with conditional variance $h_t = \gamma_0 + \alpha_1 \epsilon_{t-1}^2 + \beta h_{t-1}$.
[c] Standard errors.
[d] Simulated critical values.
[e] $\chi^2(2)$ p value.
[f] $\chi^2(1)$ p value.
[g] Schwarz values for the unrestricted model.

The low-variance parameter (γ_0) is 3.55×10^{-3}, implying an asymptotic variance of 4.15×10^{-3} for a long but finite realization that remained in the low-variance state. The high-variance parameter (γ_1) is 3.46×10^{-2}, implying an asymptotic variance of 4.46×10^{-2} for the high-variance state. The asymptotic variance in the high-variance state is more than 10 times as high as that in the low-variance state. The probability of staying in the low-variance state (p) is .99, and the probability of staying in the high-variance state (q) is .94. These numbers indicate that the asymptotic variance has a tendency to stay where it was in the previous period. The unconditional probabilities of being in the low-variance and the high-variance states are .83 and .17, respectively. The persistence parameter of the Markov process (η) is .93, but the sum of the ARCH parameters ($\alpha_1 + \alpha_2$) is .14, so that the residual $\{\epsilon_t\}$ is covariance stationary. The unconditional variance for the process $\{\epsilon_t\}$ is 1.17×10^{-2} from the proposition in Section 2.

As a comparison, I also report the estimates of the traditional AR(1)–ARCH(2) model and the AR(1)–GARCH(1, 1) model in Table 1. Notice that the first and second ARCH coefficients of .41 and .40 are much greater than those from the switching–AR(1)–Markov–ARCH(2) model and also statistically significant. The sum of the GARCH coefficients ($\alpha_1 + \beta$) is .98. Both the ARCH(2) and GARCH(1, 1) estimates indicate the strong persistence of the volatility process.

Figure 2 plots the estimated conditional variance for the AR(1)–ARCH(2) model. The picture shows a pattern of dramatic changes in the conditional volatility. Figure 3 plots the conditional variances for the switching–AR(1)–Markov–ARCH(2) model. Because the states, S_t, are not observable, I calculate expected value of ϵ_t^2 conditional on $r_{t-1}, \ldots, r_1$ as

$$
\begin{aligned}
&E\left(\epsilon_t^2 \mid r_{t-1}, \ldots, r_1\right) \\
&\quad = \sum_{S_t, \ldots, S_{t-m}} E\left[\epsilon_t^2 \mid r_{t-1}, \ldots, r_1, S_t, \ldots, S_{t-m}\right] \\
&\qquad \times p(S_t, \ldots, S_{t-m} \mid r_{t-1}, \ldots, r_1) \\
&\quad = \sum_{S, \ldots, S_{t-m}} \left(\gamma_0 + \gamma_1 S_t + \sum_{i=1}^{g} \alpha_i \epsilon_{t-1}^2\right) \\
&\qquad \times p\left(S_t \mid S_{t-1}\right) p\left(S_{t-1}, \ldots, S_{t-m} \mid r_{t-1}, \ldots, r_1\right).
\end{aligned} \tag{6}
$$

Interestingly, the data allow for clearly distinguishing be-

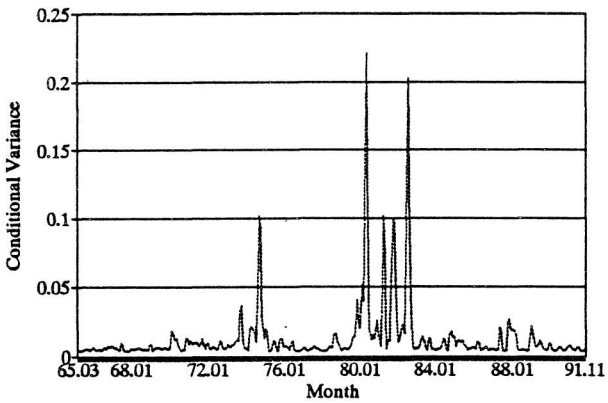


Figure 2. Conditional Variance, AR(1)–ARCH(2): 03/65–11/91.

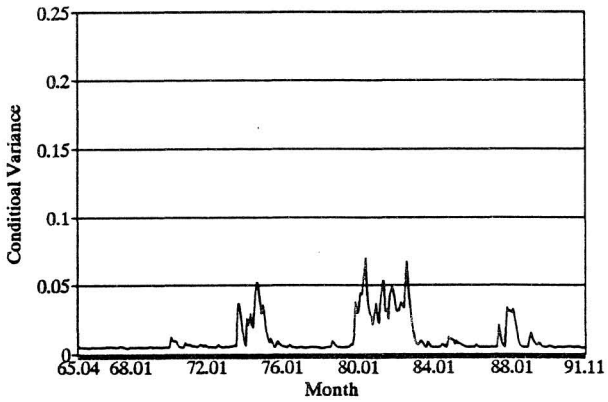


Figure 3. Conditional Variance, Switching–AR(1)–Markov–ARCH(2): 04/65–11/91.

tween the two regimes. I also find the conditional variances in the low-variance state to be surprisingly smooth. Note also that Figures 2 and 3 are on the same scale.

3.3 Some Diagnostics

3.3.1 Likelihood-Ratio Tests. To assess the performance of the switching–AR(1)–Markov–ARCH(2) model, I first test the model against the traditional AR(1)–ARCH(2) model. The traditional AR(1)–ARCH(2) model is a special case of the switching–AR(1)–Markov–ARCH(2) model when $c_1 = \gamma_1 = 0$. When c_1 and γ_1 are 0, however, the transition probabilities of the Markov process are not identified under the null, as is the case with essentially all Markov-switching models. In this case, the classical test statistics do not have standard distributions. Davies (1977) suggested a procedure based on Roy's union-intersection principle. He also provided an upper bound on the significance level. Watson and Engle (1985) employed an approximation to Davies's method to test for regression parameter constancy and provide some Monte Carlo results. Here, I perform a small Monte Carlo experiment to provide information on the distribution of the likelihood-ratio statistic. I replicate 100 data series with a sample size of 328 according to the estimates from a traditional AR(1)–ARCH(2) model as reported in Table 1. Then for each generated series, I fit a switching–AR(1)–Markov–ARCH(2) model and calculate the likelihood-ratio statistic. The mean and the standard deviation of the simulated likelihood-ratio statistics are 3.14 and 4.73. The empirical 90%, 95%, and 99% critical values from the simulations are 8.86, 10.39, and 15.40, respectively. For the actual data, the likelihood values for the switching–AR(1)–Markov–ARCH(2) model and the AR(1)–ARCH(2) model are 339.27 and 319.21. The likelihood-ratio statistic is 2(339.27–319.21) = 40.12. Thus the traditional AR(1)–ARCH(2) model can be rejected at a 1% level of significance.

The second diagnostic test is against the null hypothesis of no ARCH effects: $H_0 : \alpha_1 = \alpha_2 = 0$. Under this null hypothesis, the model reduces to Hamilton's switching-regime model. The estimation results are reported in Table 1. The likelihood-ratio statistic is 2(339.27–337.28) = 3.98. The corresponding $\chi^2(2)$ p value is .14. Because the alternative is a one-sided hypothesis, however, the true p value will be smaller than .14. For linear inequality restrictions on the parameters of linear econometric models, Wolak (1989) showed that the null distribution of the likelihood-ratio statistic is a weighted sum of χ^2 distributions. In our case, we may therefore easily reject the null of no ARCH effects. In any event, the result seems to indicate that the ARCH effects in the monthly interest-rate data are much less strong than previously believed. On the other hand, the regime shifts in the asymptotic variance have to be taken into account if we want a completely satisfactory description of the data.

The third diagnostic computed is a test for structural changes in the conditional mean: $H_0 : c_1 = 0$. The estimates under this null are also reported in Table 1. The likelihood-ratio statistic has a $\chi^2(1)$ value of 2(339.27–338.31) = 1.92, with a p value of .17. I therefore cannot reject the null of no structural change in the conditional mean of the excess return series.

Finally, I test the hypothesis of no structural change in the constant intercept in the ARCH specification: $H_0 : \gamma_1 = 0$. Estimates from this switching–AR(1)–ARCH(2) model are reported in the sixth column of Table 1. Not surprisingly the ARCH coefficients are now significant. The likelihood value is 335.17, however, providing a $\chi^2(2)$ statistic of 2(339.27=-335.17) = 8.20. The corresponding p value is smaller than .01. I therefore reject the null of no structural change in the constant intercept.

I conclude from the likelihood-ratio tests discussed previously that the most important feature of the monthly excess returns on the three-month T-bill is the structural change in the constant intercept in the ARCH specification. I find weak evidence of structural change in the mean of the excess return series, as well as weak evidence of ARCH effects.

3.3.2 Residual Diagnostics. In this section I perform some residual diagnostics. Including structural changes in the conditional mean will render the innovation ϵ_t unobservable, however, which is a problem in calculating the normalized residuals. I restrict the attention to an AR(1) model for the mean process and carry out the residual diagnostics for the ARCH(2), GARCH(1, 1), Markov–ARCH(2), and Hamilton's original switching-variance models. The means

Table 2. Residual Diagnostics

Statistics	ARCH(2)	GARCH(1, 1)	Markov–ARCH(2)	Switching variance
Mean of $\epsilon_t/h_t^{1/2}$	.04	.04	.03	.02
Mean of ϵ_t^2/h_t	1.00	.98	.93	.93
Skewness	.88	.72	.71	.67
Kurtosis	5.24	4.37	4.03	3.98
$Q_1(10)$ for $\epsilon_t/h_t^{1/2}$	12.83	11.69	10.70	9.68
	(.17)[a]	(.23)	(.30)	(.38)
$Q_1(20)$ for $\epsilon_t/h_t^{1/2}$	30.43	27.36	25.76	24.20
	(.05)[a]	(.10)	(.14)	(.19)
$Q_2(10)$ for ϵ_t^2/h_t	20.82	6.76	9.05	7.61
	(.02)[b]	(.75)	(.53)	(.67)
$Q_2(20)$ for ϵ_t^2/h_t	36.86	17.34	21.81	16.01
	(.01)[b]	(.63)	(.35)	(.72)

[a] $\chi^2(9)$ and $\chi^2(19)$ p values.
[b] $\chi^2(10)$ and $\chi^2(20)$ p values.

for the standardized residuals and squared standardized residual, as well as the coefficients of skewness, kurtosis, and the modified Box–Pierce tests for the 10 and 20 serial correlations in both the standardized residuals and squared standardized residuals, are reported in Table 2. For the Markov–ARCH(2) model and the switching-variance model, the residual is normalized by $E(\epsilon_t^2 \mid r_{t-1}, \ldots, r_1)$ calculated similarly to that described in Equation (6).

It is noteworthy that the ARCH(2) model for the conditional variance does not provide a sufficient fit in that the null hypothesis of no serial correlation in the squared standardized residuals is rejected at conventional levels. Both the Markov–ARCH(2) and switching-variance model provided much better fit. Given that the ARCH effects are weak, it is not strange that the results from the Markov–ARCH(2) and switching-variance models are very close.

3.4 Smoothed Inference

In Figures 4 and 5 I plot the filtered and the smoothed probabilities that the residual process is in a high-variance state. These probabilities are constructed using the estimated parameters from the switching–AR(1)–Markov–ARCH(2) model. Most of these probabilities are close to either 0 or suggesting that the model is well suited at the underlying pattern of dichotomous shifts between the low- and the high-variance phases in the data. Based on the criterion $p(S_t = 1 \mid I_T) > .5$, I identify three periods that are in the high-variance state. The first high-variance period is 1973:8–1974:12, associated with the major oil shock, which lasted for 17 months. The second period is 1979:8–1982:8, associated with the Federal Reserve's policy change, which lasted for 37 months. The third and last-high-variance period occurred 1987:9–1987:11, corresponding to the 1987 stock-market crash. If the criterion is instead $p(S_t = 1 \mid I_T) > .9$, the first regime shift occurred during the 1974:2–1974:8 period and lasted for 7 months, whereas the second regime shift occurred during the 1979:9–1982:7 period and lasted for 35 months. It is not surprising that one of the regime shifts identified in this article is essentially the same as that uncovered by Hamilton (1988).

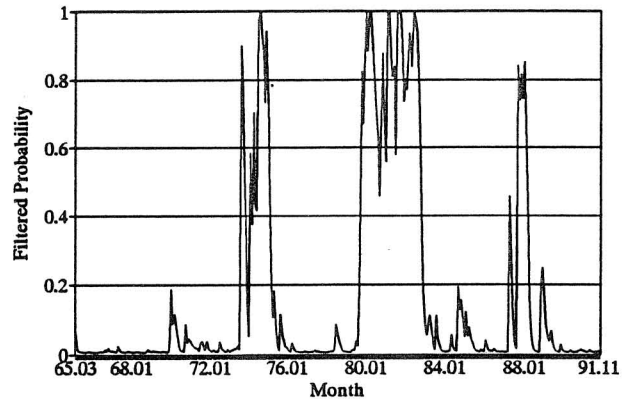


Figure 4. Filtered Probability: 03/65–11/91.

In this article, the monetary-regime shift occurred from the fourth quarter of 1979 to the third quarter of 1982 on a filtered inference. Driffill (1992) calculated Hamilton's model using data from the Chase Econometrics data base for the three-month T-bill and the Federal Reserve Board's 10-year constant-maturity government bond, quarterly, from the second quarter of 1953 to the second quarter of 1989. He found that the monetary regime shift occurred from the third quarter of 1979 to the third quarter of 1983, a result that is fairly robust. The probability of being in a different regime in the year 1974 is only around .2, however.

Prior to October 1979, the Federal Reserve used the federal funds rate as its instrument to control $M1$. In an inflationary environment such an approach often failed to achieve its targets. In an attempt to place greater emphasis on limiting inflation, the Federal Reserve switched to nonborrowed reserve targeting until October 1982. This change in the Federal Reserve's operating procedures is "one of the more dramatic events in the recent history of monetary policy" (Spindt and Tarhan 1987, p. 107). It is interesting to note that the switching ARCH model introduced here identifies the monetary regime shift with the dates of change being very close to the actual dates of change.

Marscaro and Meltzer (1983), Walsh (1984), and Melton and Roley (1990) showed that interest-rate volatility tends to

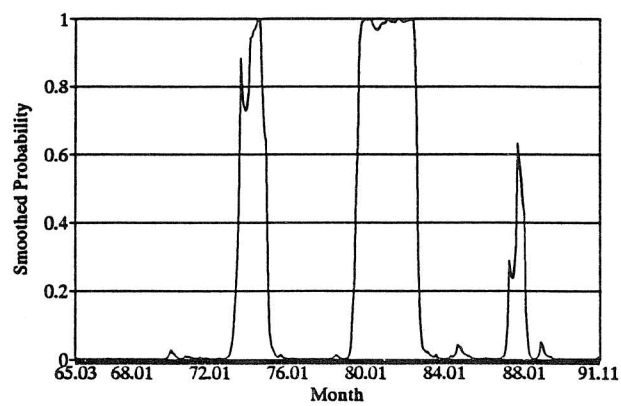


Figure 5. Smoothed Probability: 03/65–11/91.

be higher under an operative procedure that targets reserve aggregates than under one that targets nominal interest rates. Baxter (1989) showed that interest-rate volatility can increase if a change in the monetary rule induces uncertainty. The empirical findings in this article are therefore consistent with these theoretical results.

4. CONCLUSION

This article presents a new approach to modeling the variability of financial time series by combining the ARCH model developed by Engle (1988) and the Markov-switching model proposed by Hamilton (1982). This enables me to capture not only sudden dramatic changes in the underlying fundamentals or regime shifts but also time-varying conditional second moments within the regimes. My model is supported empirically with data on three-month T-bill monthly excess returns from August 1964 to November 1991. I believe that monthly excess returns for T-bills of other maturities should also support the Markov–ARCH model. More specifically, I reach the following conclusions.

First, with respect to the monthly excess returns of the three-month T-bill, the persistence issue can be resolved by taking into account sudden, discrete shifts in the asymptotic variance of the Markov–ARCH process. The ARCH coefficients are significantly reduced, and the ARCH process is much less persistent. In contrast, previous studies in general find that the ARCH process is highly persistent.

Second, I identified two periods during which there is a regime shift, the 1974:2–1974:8 period associated with the oil shock and the 1979:9–1982:8 period associated with the Federal Reserve's policy change. I find that the asymptotic variance of the Markov–ARCH process for the continued realizations in these two episodes is more than 10 times as high as that of the other periods. My results suggest that regime shifts have more effect on the properties of the data over the entire sample period, and I cannot reject the null that there are no ARCH effects in the volatility process based on a standard likelihood-ratio test.

As for other implications of these results, it is not clear whether more careful models of the volatility for the T-bill returns and other interest-rate-related asset returns will produce significantly different outcomes in asset-pricing models that rely on time-varying variances and covariances. Moreover, in option pricing, the valuation of the option depends crucially on the evolution of the volatility process. Finally, the findings in this article have important implications for optimal hedge ratios that involve functions of the conditional variances and covariances. Cecchetti, Cumby, and Figlewski (1988) employed a bivariate linear ARCH(3) model to estimate the conditional variances and covariance for monthly holding returns of 20-year treasury bonds and treasury-bonds futures. They found substantial variation in the estimates for the risk-minimizing hedge ratios and the expected utility-maximizing hedging ratios for an investor with a logarithmic utility function. Their results should be carefully interpreted because the data also involve the 1979:10–1982:10 monetary experiment period. One may still expect to see substantial variation in the hedge ratios with a more careful volatility process, but the pattern of the variation could be very different.

For possible extensions of the article, one may naturally consider a Markov–ARCH–in–MEAN model. Including lagged residuals in the conditional mean, however, gives rise to formidable complications in computation because the conditional density will then depend on the entire history of the states. This is essentially the same problem as in the Markov–GARCH model discussed previously. Empirically high-frequency exchange-rate data also show very pronounced persistence, motivating the integrated GARCH model. Because monetary policy changes will undoubtedly have great impact on exchange-rate dynamics, it would be interesting to see whether exchange-rate data also support the Markov–ARCH model and whether the persistence issue can be similarly resolved with respect to such data. This is left for future research.

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APPENDIX: PROOF OF THE PROPOSITION

Let $M_t = (\epsilon_t^2, \ldots, \epsilon_{t-g}^2)'$, $\delta = (1, 0, \ldots, 0)'_{1\times g}$, and

$$\Delta = \begin{pmatrix} \alpha_1 & \cdots & \alpha_g & 0 \\ 1 & \cdots & 0 & 0 \\ \cdot & \cdots & \cdot & \cdot \\ \cdot & \cdots & \cdot & \cdot \\ 0 & \cdots & 1 & 0; \end{pmatrix}_{(g+1)\times(g+1)};$$

then $E_{t-j}(M_t) = (\gamma_0 + \gamma_1 E_{t-j}S_t)\delta + \Delta E_{t-j}(M_{t-1})$. Taking successive expectations and using the fact $E_{t-j}S_t = \pi + \eta^j(\pi_{t-j} - \pi)$, $\pi_{t-j} = E_{t-j}S_{t-j}$ yields

$$E_{t-j}(M_t) = (\gamma_0 + \gamma_1\pi)(I + \Delta + \cdots + \Delta^{j-1})\delta + \Delta^j M_{t-j} + \gamma_1(\pi_{t-j} - \pi)(\eta^j + \eta^{j-1}\Delta + \cdots + \eta\Delta^{j-1})\delta. \quad \text{(A.1)}$$

The term of interest in (A.1) can be written as

$$(\eta^j I + \eta^{j-1}\Delta + \cdots + \eta\Delta^{j-1}) = M(\eta^j I + \eta^{j-1}J + \cdots + \eta J^{j-1})M^{-1}, \quad \text{(A.2)}$$

where $\Delta = MJM^{-1}$ is the Jordan decomposition of Δ. But

$$\eta^j I + \eta^{j-1}J + \cdots + \eta J^{j-1} = (\eta^j I + \eta^{j-1}J + \cdots + \eta^{j-l}J^l) + (\eta^{j-l-1}J^{l+1} + \cdots + \eta J^{j-1}). \quad \text{(A.3)}$$

Notice that

$$\begin{aligned}&\left(\eta^{j-l-1}J^{l+1}+\cdots+\eta J^{j-1}\right)\\&\quad=J^{l}\left(\eta^{j-l-1}J^{1}+\cdots+\eta J^{j-1-l}\right)\\&\quad\leq J^{l}(J^{1}+J^{2}+\cdots),\end{aligned}\tag{A.4}$$

where $\leq$ is an element-by-element inequality. If the roots of Equation (5) are all outside the unit circle, then the diagonal elements of J are all inside the unit circle. Hence for any $\delta > 0$ there exists a $l > 0$ such that every element of the matrix in (A.4) is less than $\delta/2$ for every $j > l + 1$. Similarly, the first term on the right side of (A.3) is bounded by

$$\begin{aligned}&\left(\eta^{j}I+\eta^{j-1}J+\cdots+\eta^{j-l}J^{l}\right)\\&\quad=\eta^{j-l}\left(\eta^{l}I+\eta^{l-1}J+\cdots+J^{l}\right)\\&\quad\leq\eta^{j-l}(I+J+J^{2}+\cdots).\end{aligned}\tag{A.5}$$

For any $\delta > 0$ and any $l > 0$, there exists a $j > 1$ such that (A.5) is less than $\delta/2$. Hence there exists a j for which every element of (A.3) is less than δ, establishing that (A.2) converges to zero matrix as j does to infinity.

The second term vanishes as long as the roots of Equation (5) are all outside the unit circle, while the first term in (A.1) converges in probability: $p\lim_{j\to\infty} E_{t-j}(M_t) = (\gamma_0+\gamma_1\pi)(I-\Delta)^{-1}\delta$. Because this does not depend on initial condition or on t, this is the common variance for all t. Each element of the preceding vector can be written as

$$E(\epsilon_t^2)=\frac{\gamma_0+\pi\gamma_1}{1-\sum_{i-1}^{g}\alpha_i}.$$

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